

Integrated Circuit ECE326

Lec. 11

CMOS Processing Technology

Dr. Ahmed Reda Mohamed^{1,2}

¹Ph.D, the MoE Key Laboratory of Artificial Intelligence & Bio-Circuits and Systems Laboratory (*BiCASL*), Department of Micro-Nano Electronics, Shanghai Jiao Tong University, Shanghai, China.

²Assistant professor, Department of Electronics and Communications, Zagazig University, Zagazig , Egypt.

Outlines



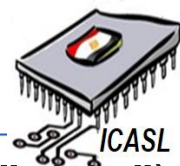
- ☐ What's On a Chip
- ☐ Construction of MOSFET
- ☐ Flow of IC process
- ☐ IC Masks
- ☐ Fabrication Process of CMOS Inverter

Outlines

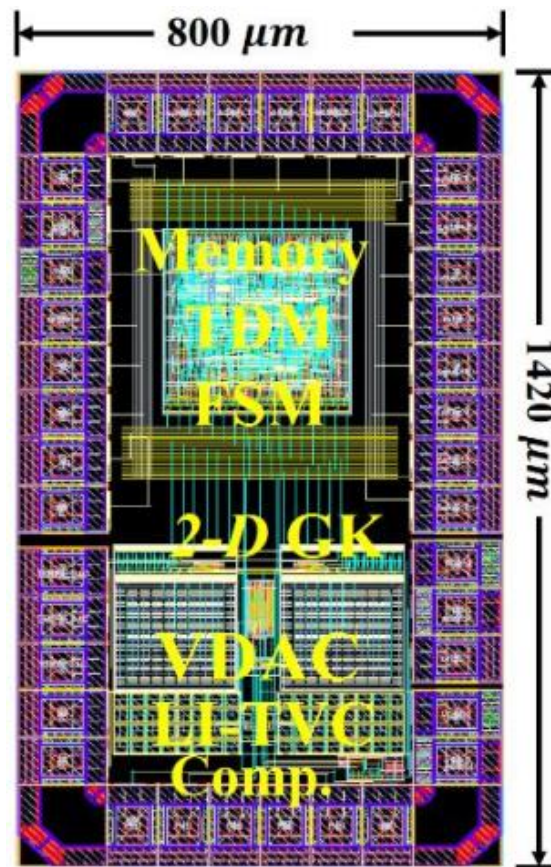
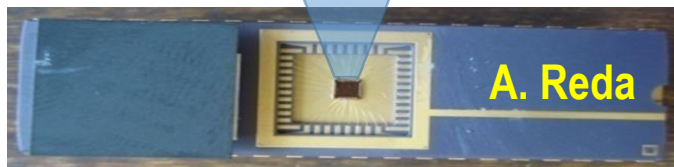
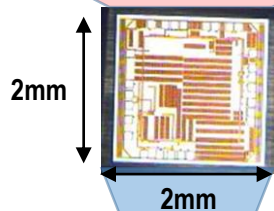
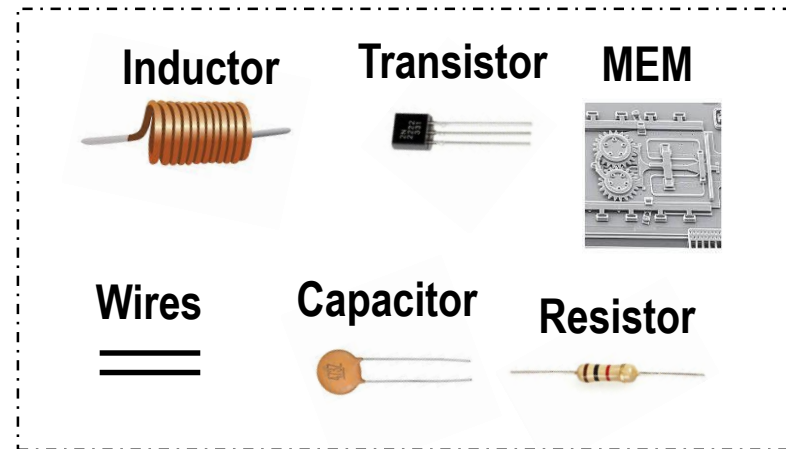


- ☐ **What's On a Chip**
- ☐ **Construction of MOSFET**
- ☐ **Flow of IC process**
- ☐ **IC Masks**
- ☐ **Fabrication Process of CMOS Inverter**

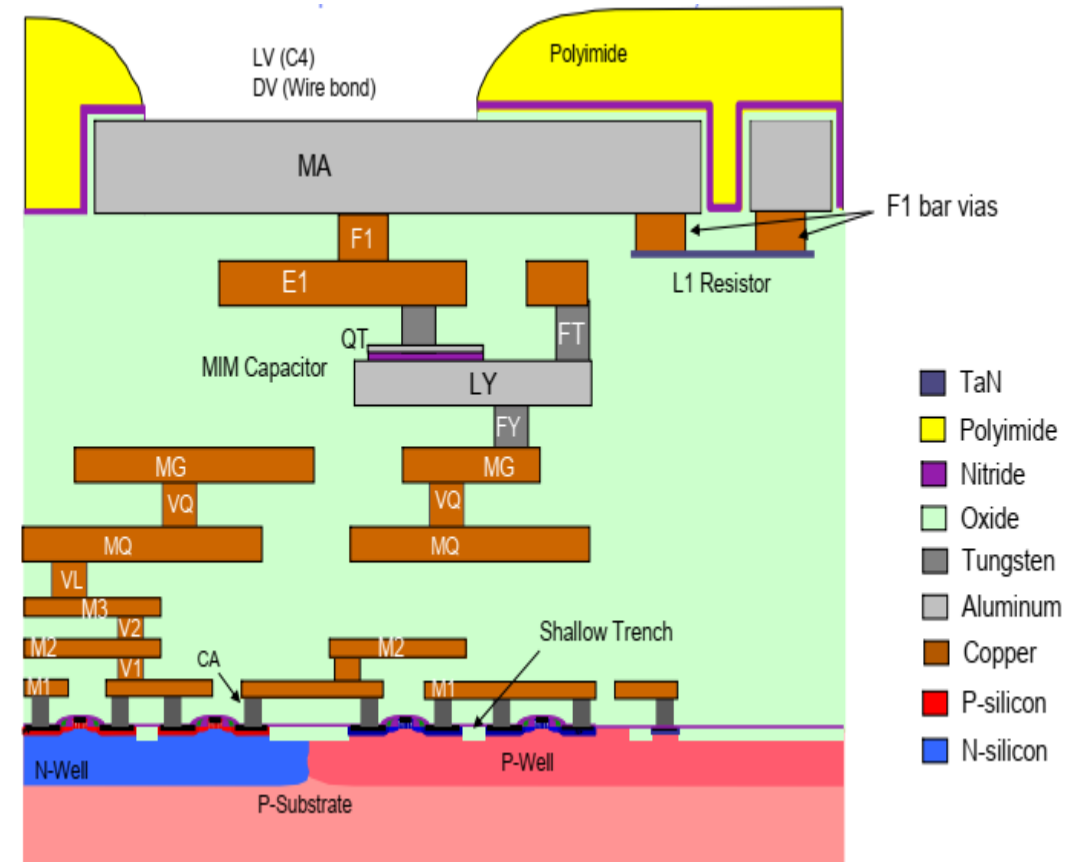
What's On a Chip



❑ An **integrated circuit (IC)** is a set of electronic circuits on one small flat piece (or "chip") of semiconductor material, usually silicon.



Layout using CAD tool



- TaN
- Polyimide
- Nitride
- Oxide
- Tungsten
- Aluminum
- Copper
- P-silicon
- N-silicon

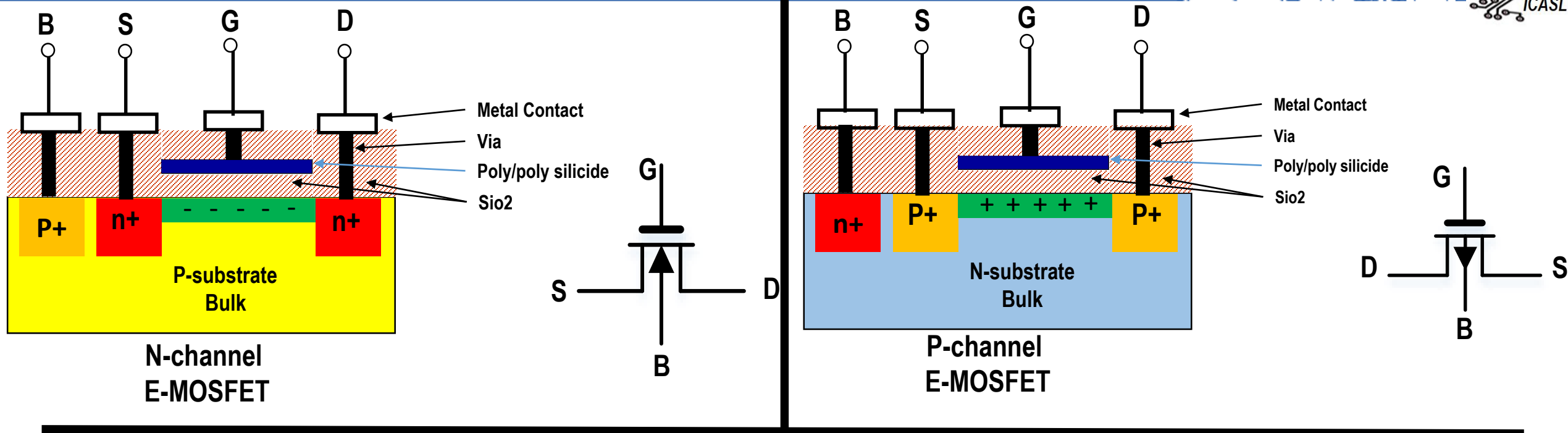
Process Cross-Section
8 layers

Outlines



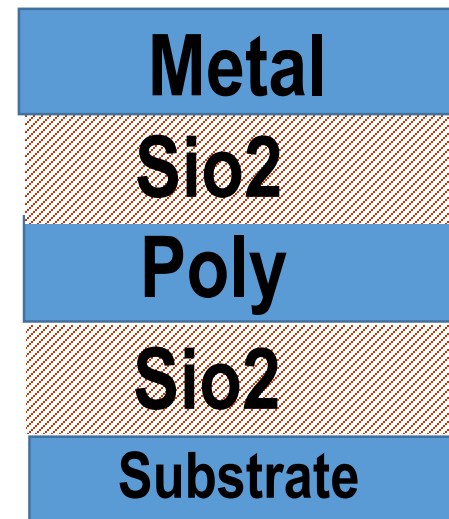
- ❑ What's On a Chip
- ❑ **Construction of MOSFET**
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Construction of MOSFET



Note:

- ❑ n+ and p+ (semi-conductor materials) are **inside** the substrate
- ❑ Poly/poly silicide and metal are **on** substrate
- ❑ SiO2 is used as an **isolation layer** between two different layers

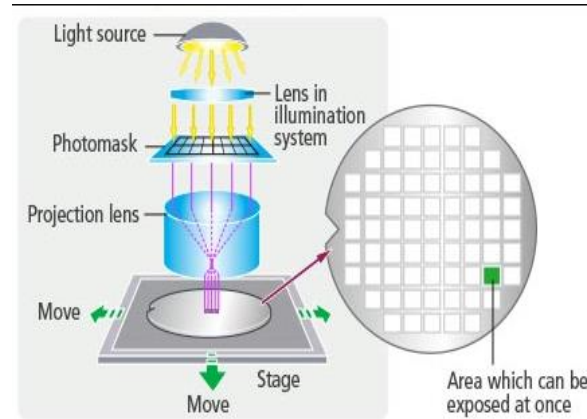


Outlines



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Flow of IC process



Sand/Silicate

Crystal preparation

Wafer preparation

Epitaxy growth

Oxidation

Photolithography

Masks

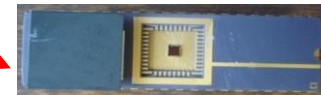
Etching

Diffusion or ion-implantation

Impurity Doping

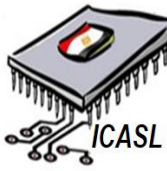
Metallization

Packaging



- ❑ Chips are built in huge factories called fabs (fabrication plants). Fabs contain “Clean Rooms”
- ❑ No dust particles are allowed to exist in the clean room, as the particle is larger than the transistor size.
- ❑ A fab processing 300 mm wafers' diameter in a 45 nm process costs about \$3 billion (that's why many semiconductor companies are fabless, like Xilinx, Qualcomm)
- ❑ Big players : Intel, Samsung, TSMC (fab only), Global Foundries, XFAB, IBM.
- ❑ Clean room in Egyptian universities (Zewail, AUC) for research.

Crystal and Wafer Preparation Process

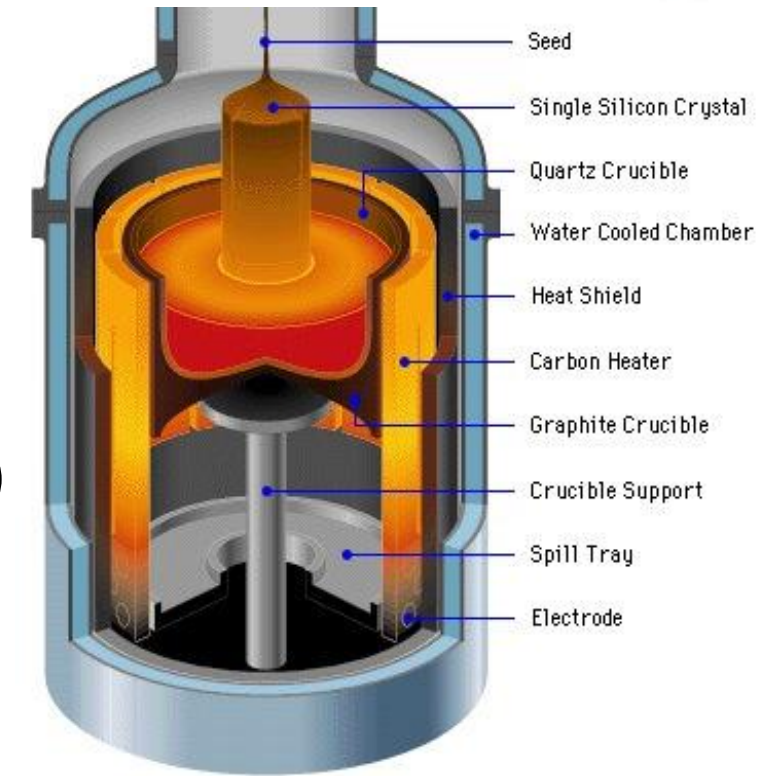


Definition:

- ❑ It is the first step to generate a wafer by **Czo-chralski** technique

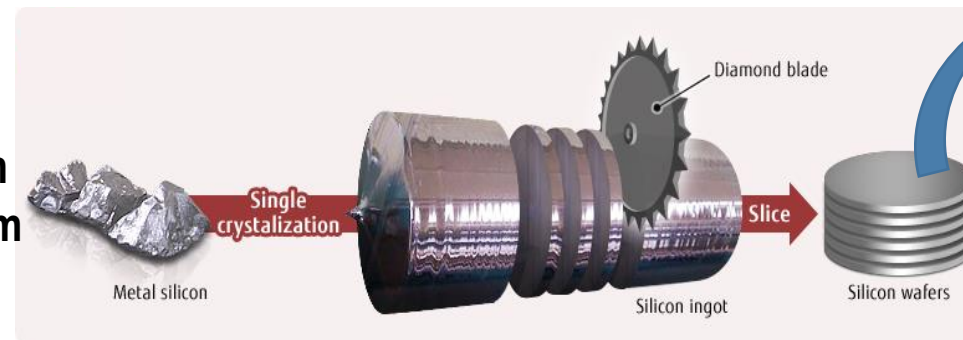
Operation:

- ❑ The source of crystal **silicon** ingot is sand/silicate
- ❑ **Heating** this materials up-to 1420 °C to provide a molten silicon
- ❑ **Doped atoms** are added to produce the n/p substrate
- ❑ To get n substrate by doping **pentavalent** atoms, Phosphorous (P)
- ❑ To get p substrate by doping **trivalent** atoms, Boron (B)
- ❑ Rotate the **seed** crystal to get a **uniform** silicon **ingot**
- ❑ Begin to cut the silicon **ingot** into **wafers**
- ❑ **Later**, each wafer would be cut into many **Dies**.

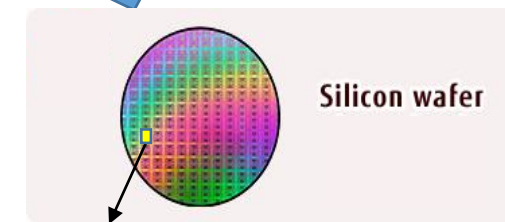


Silicon ingot dimension:

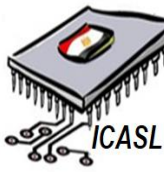
- ❑ The length would reach 2m
- ❑ The diameter would reach 100 mm: 300 mm
- ❑ The thickness would reach 250μm :1000 μm



Later

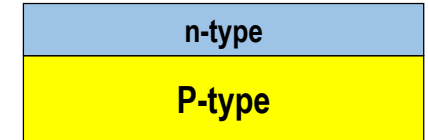


Epitaxy Growth Process



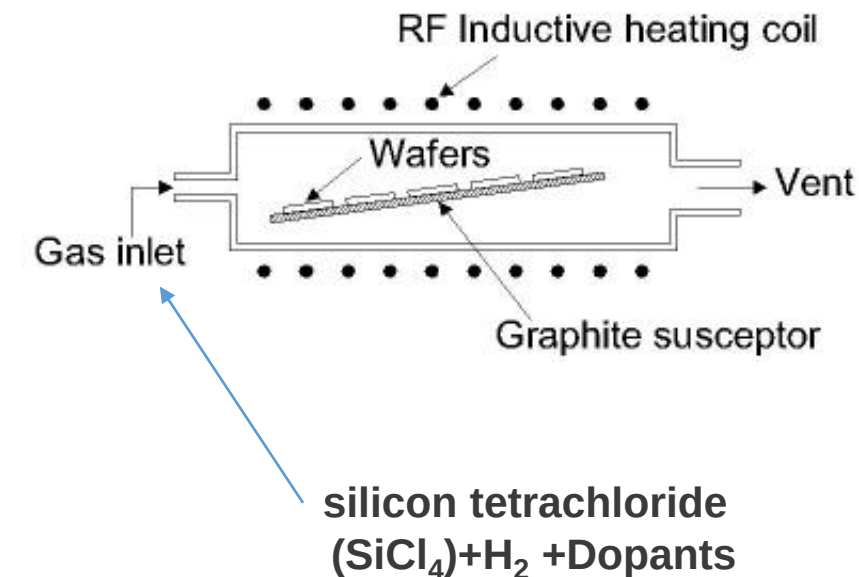
Definition:

- ❑ It is a method to re-arrange atom on the n/p substrate. It is use to grow a n-type layer on a p-type. Layer and vice versa.



Operation:

- ❑ In chemical Vapor Deposition (**CVD**), the film is formed on the surface of the substrate by thermal decomposition or the reaction of various gaseous compounds.
- ❑ The epitaxial layer formed from the gaseous vapor phase is called Vapor Phase Epitaxy (**VPE**)
- ❑ The process of CVD growth takes place in the reaction chamber with RF induction coil wound on it.
- ❑ The silicon wafers are supported by Graphite susceptor.
- ❑ The susceptor is heated by RF induction to temperature between 1100°C to 1300°C
- ❑ The doping is done by using hydrides of arsenic, boron, phosphorus. These are Arsine (AsH_3), diborane (B_2H_6) and phosphine (PH_3). In general, the film grows at 1 $\mu\text{m}/\text{min}$ typically.



Oxidation Process (1/2)



Definition:

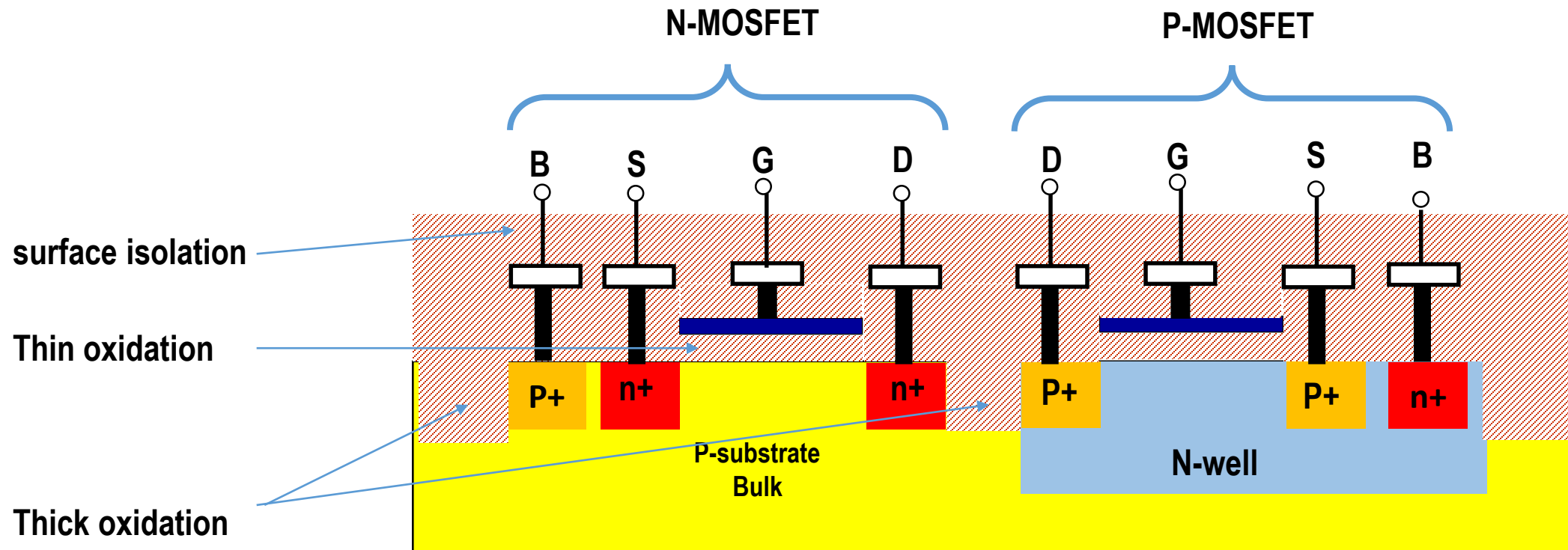
❑ It is a process to form **thin** or **thick** layer of SiO_2 on the wafer's surface.

❑ Uses

❑ To form a **thin** oxidation layer under the gate of the MOSFET .

❑ To form a **thick** oxidation layer (Field oxidation= FOX) as barrier between two transistors

❑ To form a **surface isolation** layer of the wafer's surface.

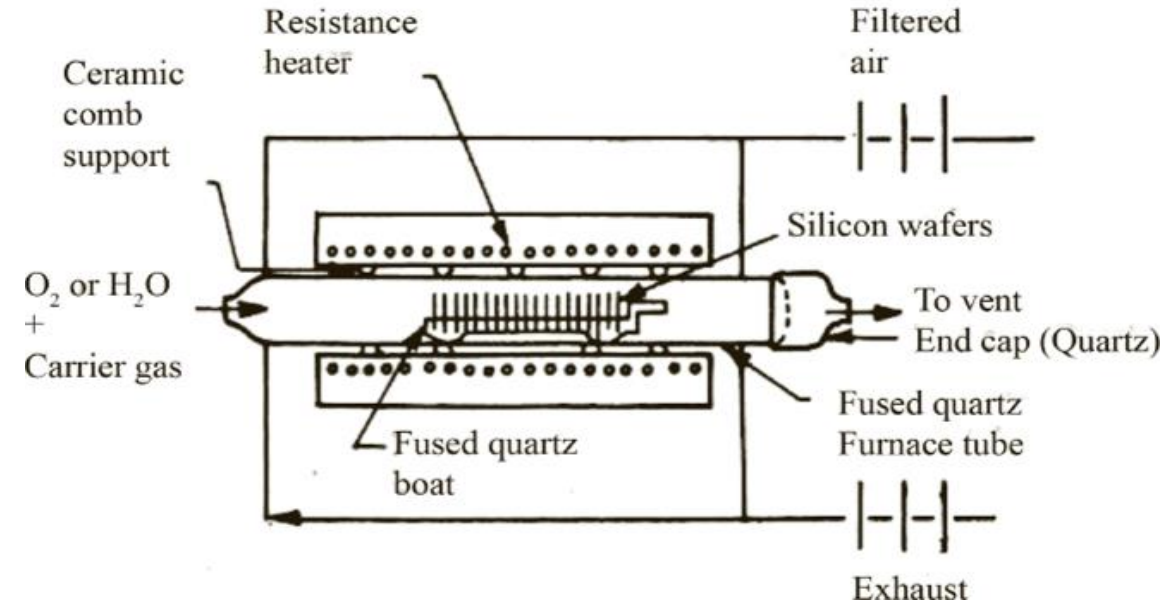


Oxidation Process (2/2)



❑ Operation

- ❑ The wafers are exposed to a source of either pure dry oxygen or pure water vapor.
- ❑ The furnace temperature is maintained between 900°C and 1200°C.



Wet Oxidation (H₂O)

- ❑ To produce a **thick** layers
- ❑ By using a water vapor at (900 °C:1000 °C)
- ❑ Used to form a **thick** oxidation layer between two transistor

Dry Oxidation (O₂)

- ❑ To produce a **thin** layer
- ❑ By using a dry oxygen at **1200 °C**
- ❑ Used to form a **thin** oxidation layer under the gate of MOSFET

Lithography Process



Definition:

- ❑ It is a process of transferring patterns of geometric shapes on a mask to a thin layer of sensitive material (Photo-resist) which covers the substrate's surface.
- ❑ **Optical/photo-lithography** is widely used in IC that uses a **photo-resist**.

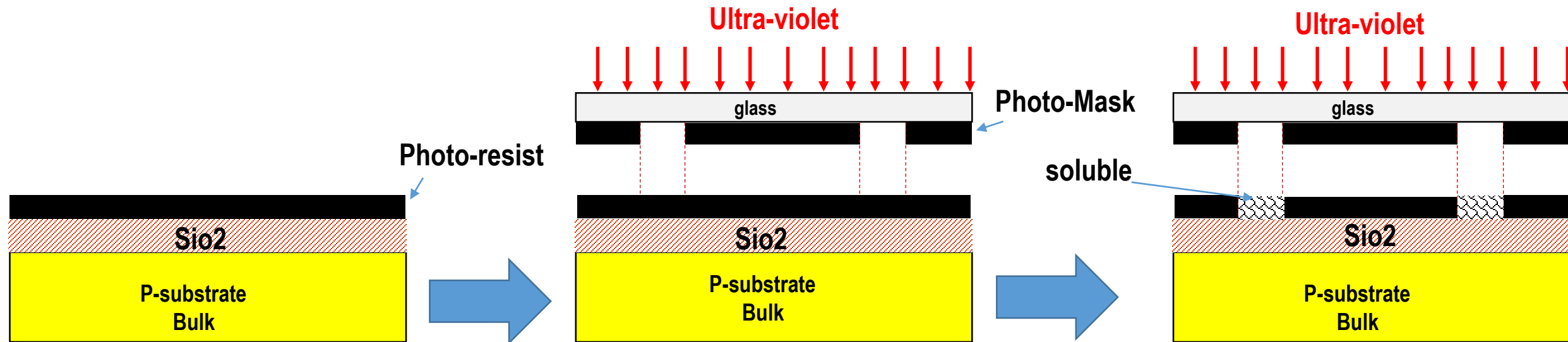


Photo-Lithography Process (1/3)



Definition:

- ☐ Lithography is a basic method of IC fabrication process.
- ☐ Process is used to transfer patterns from masks to each layer of the IC on the surface of a wafer by employing a **photosensitive**, chemically resistant layer (**photo-resist**).
- ☐ Masks are created using the layout information provided by the **designer**.
- ☐ The lithographic process is repeated on each physical layer, and the process sequence is always the same.
- ☐ The Photo-resist material becomes **soluble** (**dissolved**) when it is exposed to ultraviolet light.

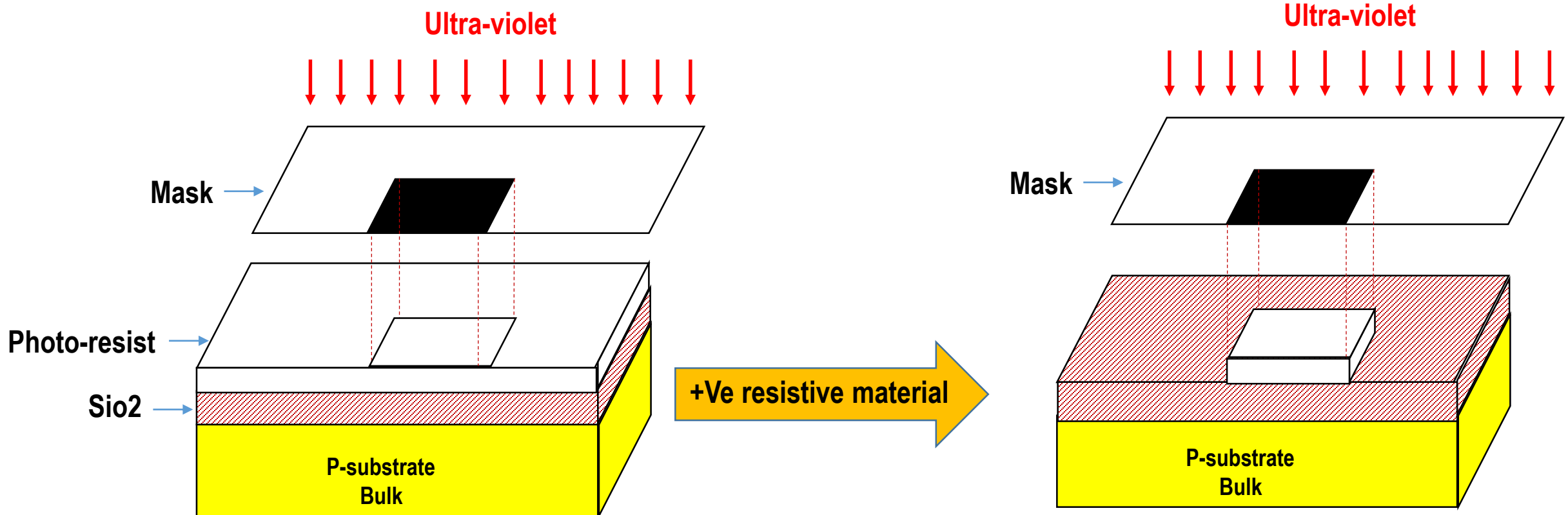
☐ Types

- ☐ Positive (+Ve) resistive material → The used material will be dissolved when it **exposes** to UV light
- ☐ Negative (-Ve) resistive material → The used material will be not dissolve when it **exposes** to UV light.

Photo-Lithography Process (2/3)



Positive (+Ve) resistive material

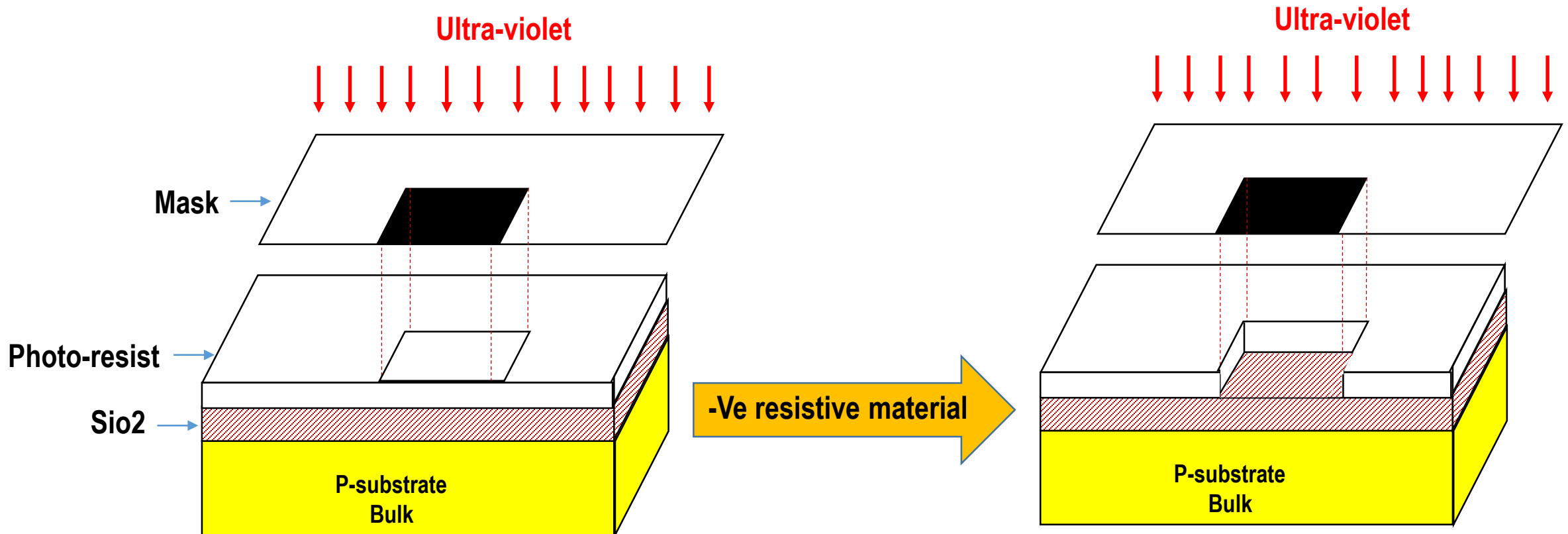


Etching process usually follows the lithography process,

Photo-Lithography Process (3/3)



Negative (-Ve) resistive material



Etching process usually follows the lithography process,

Etching Process



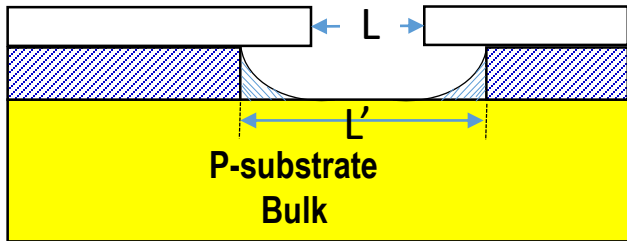
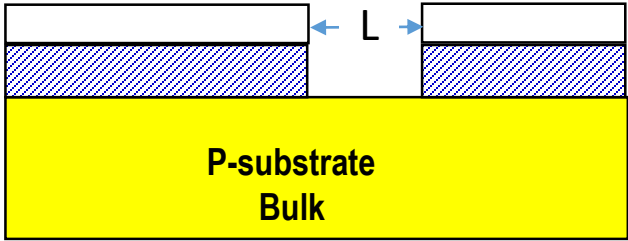
❑ Definition:

❑ It is a method to remove the undesired area and it usually follows the photo-lithography process.

❑ Etching methods

❑ Wet chemical Etching

❑ Dry Etching

Wet Chemical Etching	Dry Etching
❑ This technique uses (HF) Hydrofluoric acid ❑ HF assists in etching SiO_2 rapidly than the photo-resist layer or even Si. Disadvantages ❑ Rate of etching from 10:100 mm/min at 25 °C ❑ Rate of etching depends on:- 1. Temperature 2. Type of isolation material (SiO_2) 3. Thickness of isolation material (SiO_2) ❑ Etching occurs in all direction ❑ Etching occurs under the photo-resist.  The diagram illustrates isotropic etching. A yellow 'P-substrate Bulk' is shown at the bottom. Above it is a blue hatched layer representing the isolation material. A white photo-resist layer is on top, with a central opening of width 'L'. The etching process is shown as the blue layer is being removed from all directions (isotropically), with a dimension 'L'' indicating the width of the etched area at the substrate level.	❑ It is used to avoid the undercutting issue in the wet chemical etching ❑ Dry etching uses amount of reactant gas.  The diagram illustrates anisotropic etching. A yellow 'P-substrate Bulk' is shown at the bottom. Above it is a blue hatched layer representing the isolation material. A white photo-resist layer is on top, with a central opening of width 'L'. The etching process is shown as the blue layer is being removed vertically (anisotropically), with a dimension 'L' indicating the width of the opening at the top.

Impurity Doping Process (1/4)

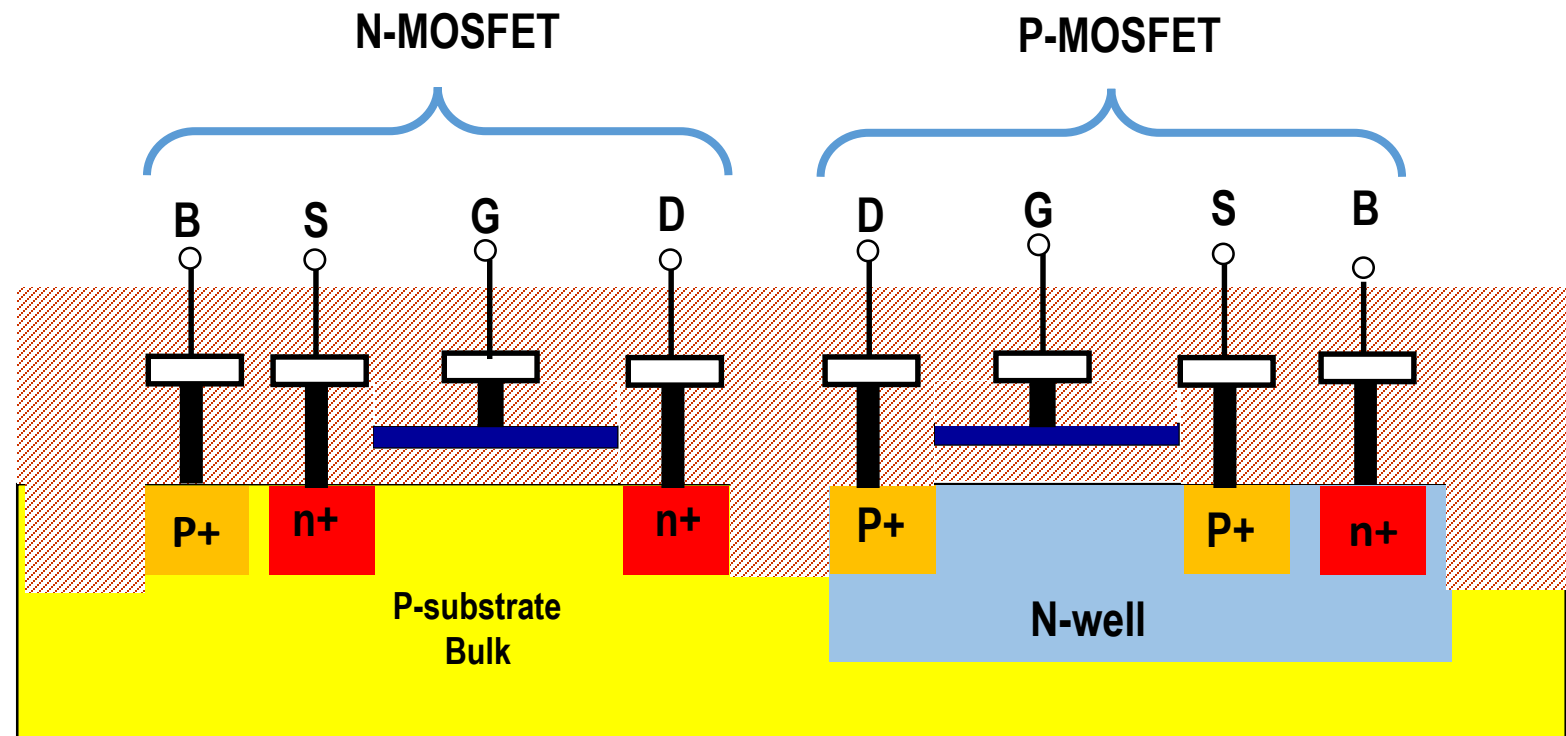


❑ Definition

- ❑ It is process to form n^+ material in p-type material or p^+ in n-type

❑ Methods

- ❑ Diffusion Process
- ❑ Ion-implantation



Impurity Doping Process (2/4)

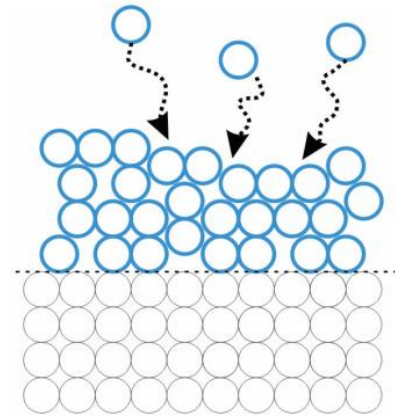


□ Diffusion Process

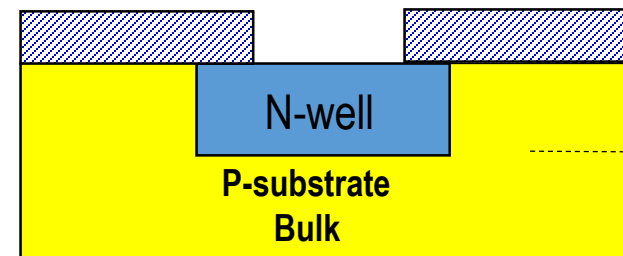
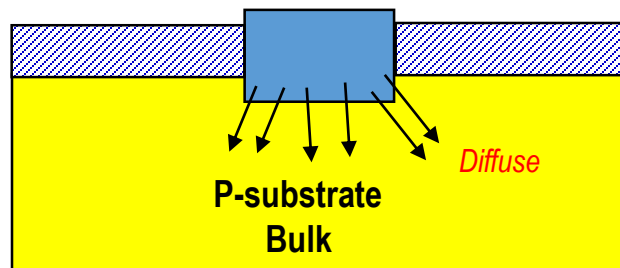
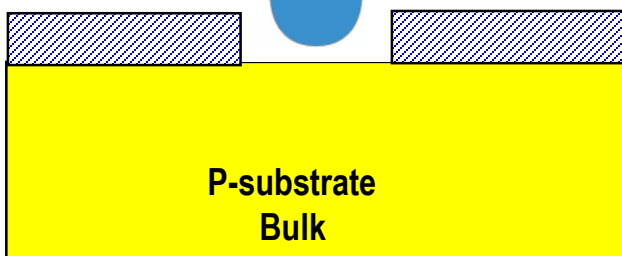
- Dopant atoms move from the surface into Si by thermal means via substitution or interstitial diffusion mechanisms. Atoms move from high concentration (on surface) to low concentration (inside Sub.)

□ Steps

- Doping material is pre-deposited on the surface of the substrate.
- The impurity atoms diffuse into the bulk of the material at Temp. 800:1200 °C
 - Penta-valent atoms are used to form n-type → phosphorus (Ph)
 - Tri-valent atoms are used to form p-type → Boron (B)



Ph

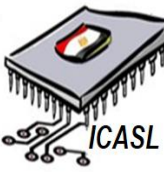


$X=0$ (Surface) → Concentration of Phs is the highest

$X=X_j$ → Concentration of Phs equals concentration of Bulk's carriers

X_j : is called Junction Depth

Impurity Doping Process (3/4)

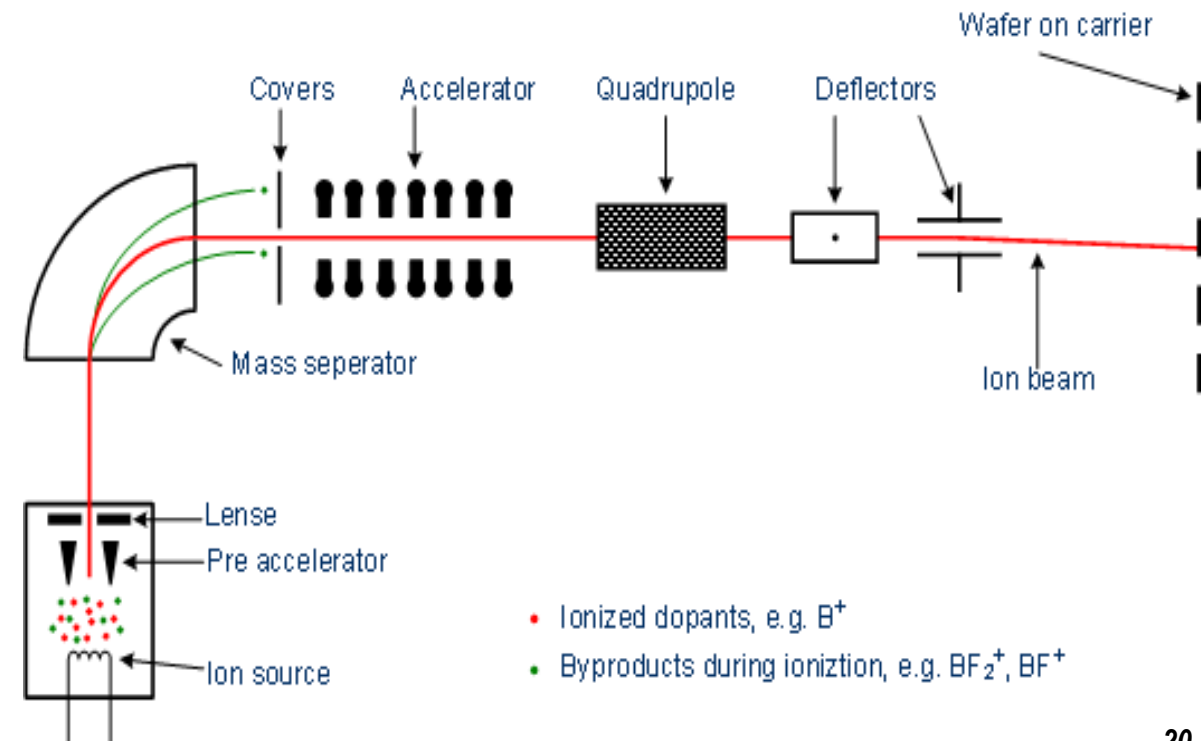
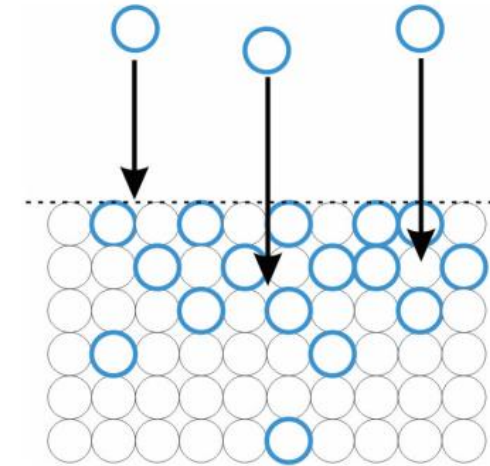


❑ Ion-implantation Process

- ❑ Dopants projected at silicon surface with selected angle/energy/dose
- ❑ Dopants penetrates silicon surface to desired depth by using ion beam injection technique.

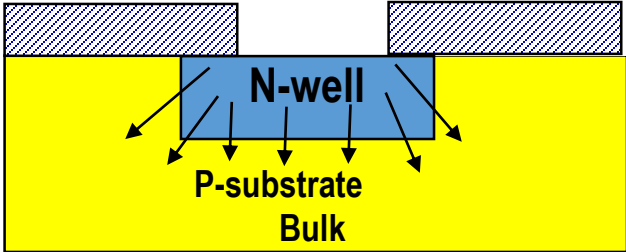
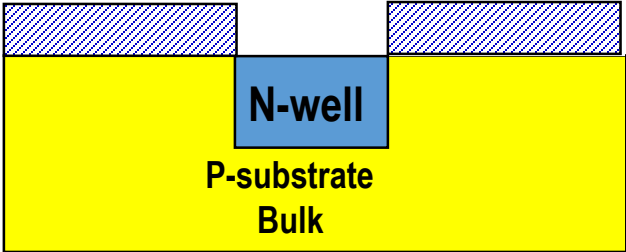
❑ Steps

- ❑ The exposed areas of the silicon wafer are bombarded with dopants with selected angle/energy/dose.
- ❑ Dopants penetrates silicon surface to desired depth ions are implanted in the silicon wafer to alter the way silicon in these areas conduct electricity.
- ❑ Ions are propelled onto the surface of the wafer at very high **velocities**.



Impurity Doping Process (4/4)



Diffusion Process	Ion implantation
Dopant atoms move from the surface into Si by thermal means via substitution or interstitial diffusion mechanisms	Dopants penetrates silicon surface to desired depth by using ion beam injection technique
Cheaper and more simplified High lateral diffusion High Temp. Cannot independently control the dopant concentration and junction depth	Expensive and more complex Less lateral diffusion Low temp. Can independently control the dopant concentration and junction depth
	

Other Processing Steps



☐ Polishing

- The surface of the wafer is polished to make it even (planer)
- Chemical mechanical polishing is often used.

☐ Metallization

- Metal layer sputtered to cover the surface of the wafer .

☐ Deposition

- Material layer deposited to cover the surface of the wafer
- Selective deposition only covers surface of selected material

Outlines



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- ☐ Flow of IC process
- ☐ **IC Masks**
- ☐ Fabrication Process of CMOS Inverter

IC Masks (1/4)



Definition:

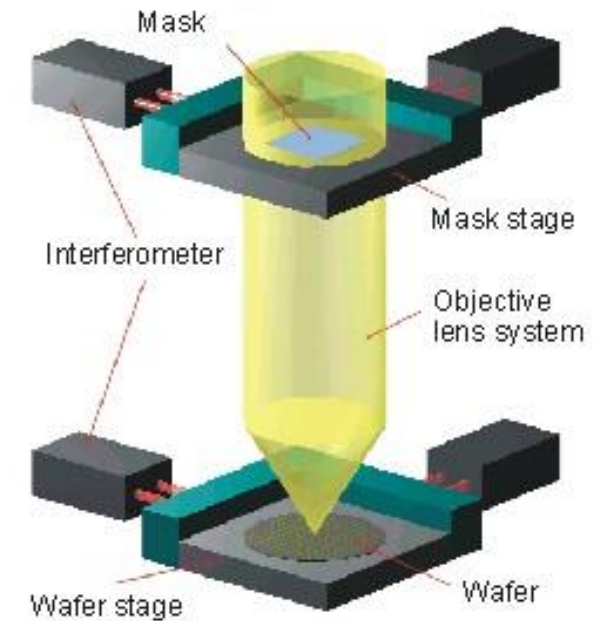
- ❑ Mask is a method in which the designed schematic circuit is coded into figure of window.
- ❑ It depends on type of the technology/process.

❑ Types of technology/process

- ❑ N-well process
- ❑ P-well process
- ❑ Twin well process
- ❑ Triple well process
- ❑ Silicon on insulator
- ❑ FinFet process

❑ Type of Masks

- ❑ Diffusion Mask
- ❑ Deposition Mask
- ❑ Connection Mask



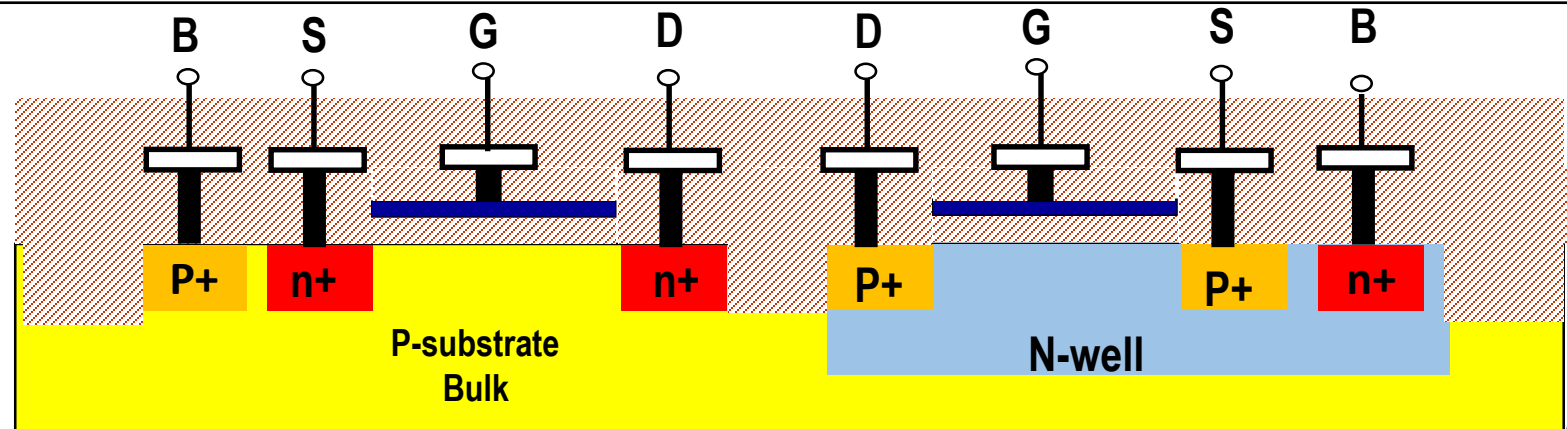
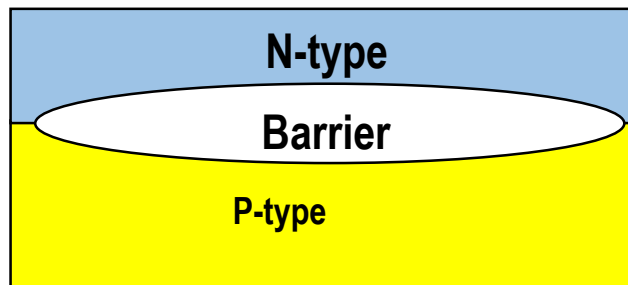
IC Masks (2/4)



□ Diffusing mask

- It is used to implement elements **into** the substrate

Mask's Type	Function
Active mask	Determine the location of transistor that will be built in the substrate
N-selected mask (n-mask)	To produce n ⁺ (drain & source) into the sub.
P-selected mask (p-mask)	To produce p ⁺ (drain & source) into the sub
n-well mask	To produce N-well (n-sub.) into P-substrate
P-well mask	To produce P-well (p-sub.) into N-substrate
Barrier-layer mask	To isolate between different transistor and martial → BJT



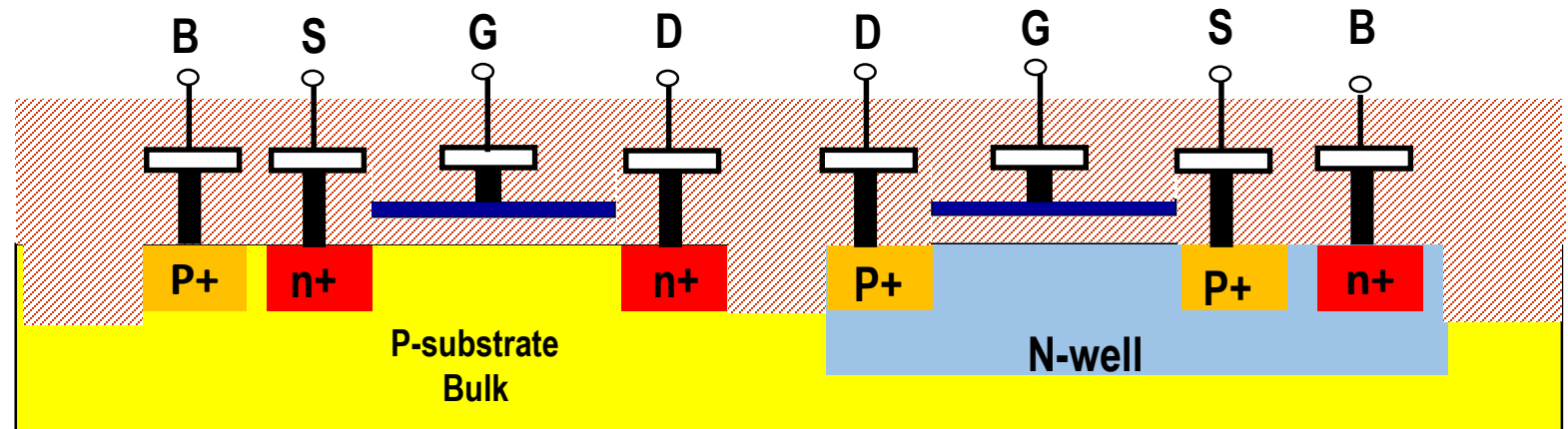
IC Masks (3/4)



□ Deposition mask

- It is used to implement element **on** the substrate

Mask's Type	Function
Poly-mask	To form the gate of the MOSFET
Metal-mask	To form a metal layer from Al or Cu which is used as a connection wire



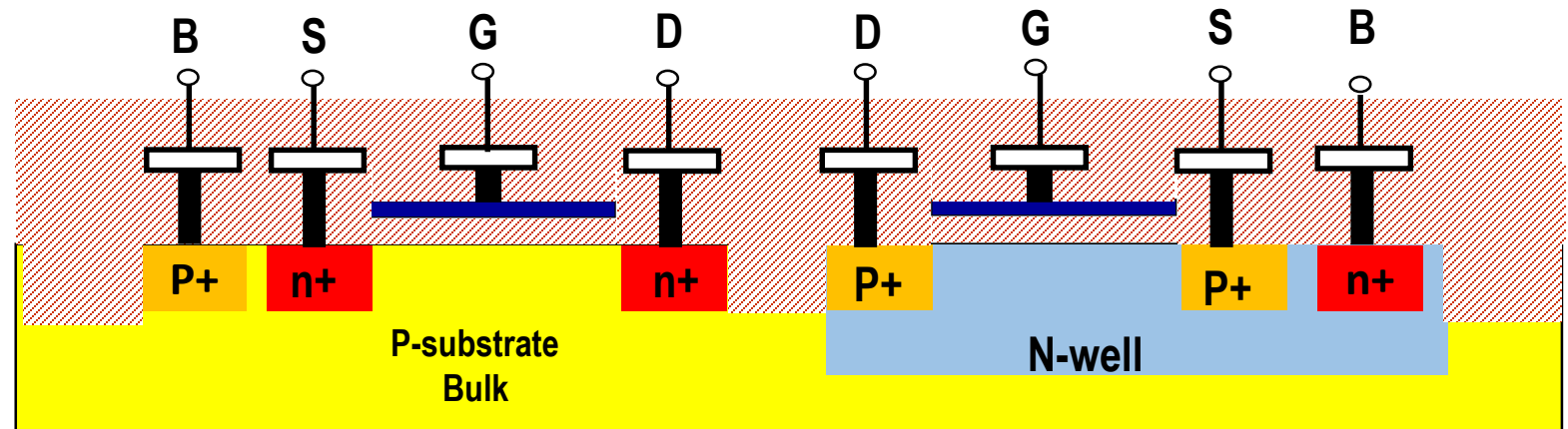
IC Masks (3/4)



❑ Connection mask

- ❑ It is used to connect between different layers

Mask's Type	Function
Active contact mask	To connect between semi-conductor and other layer
Poly-to-poly mask	To connect between two poly materials
Metal-to metal (Via) mask	To connect between two metals

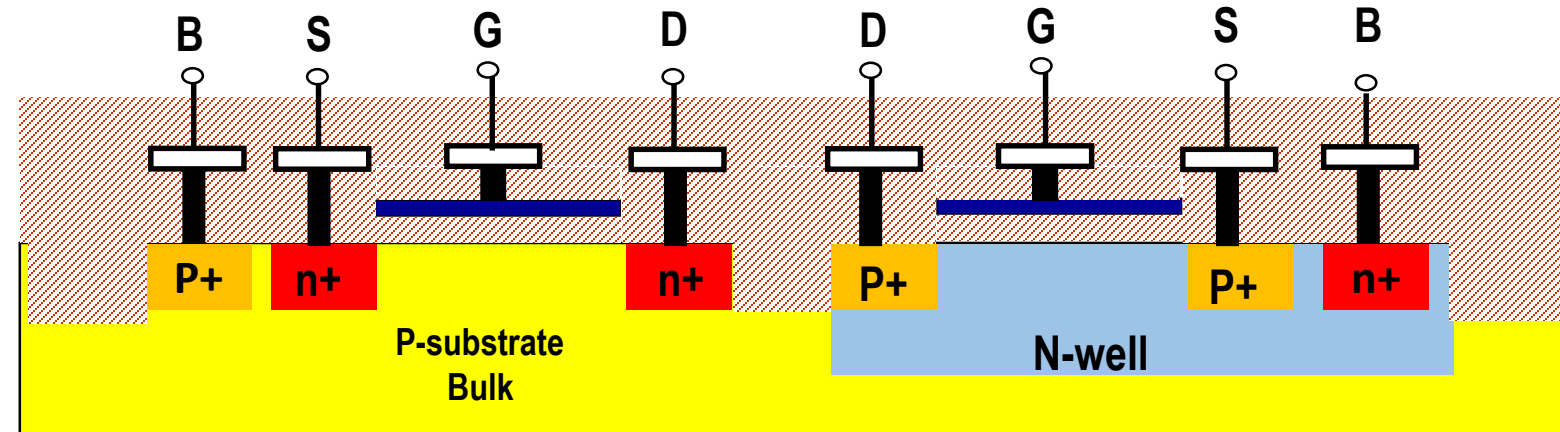
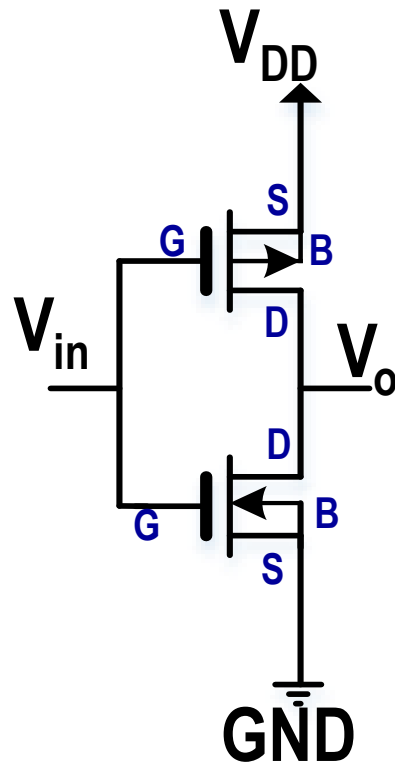
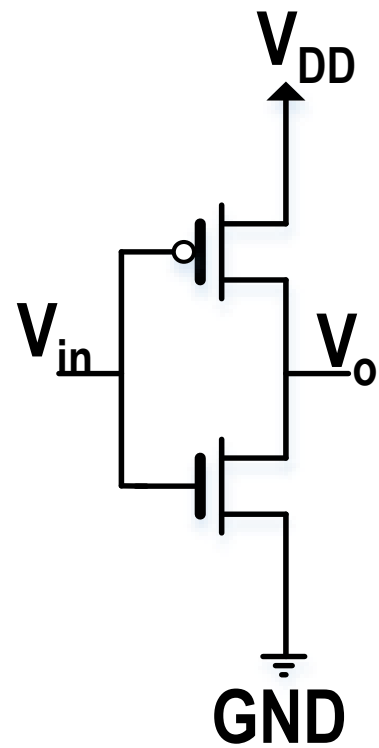


Outlines

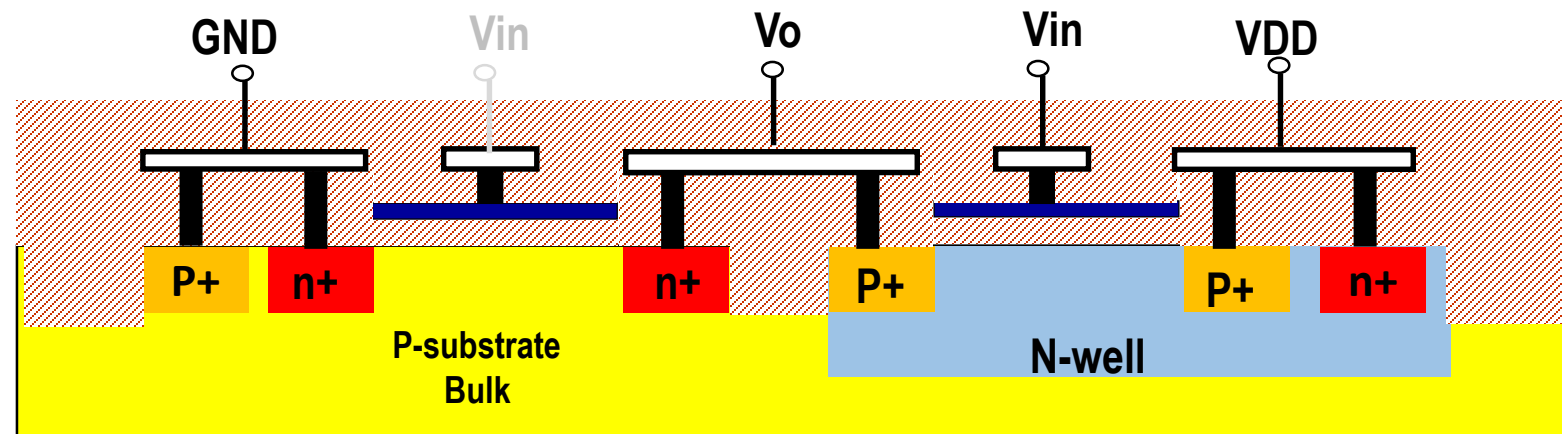


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Fabrication Process of CMOS Inverter (1/10)



N-MOSFET and P-MOSFET Transistors

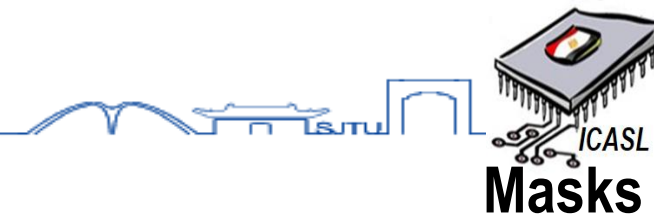


CMOS inverter

Noted:

The used technology is N-well technology

Fabrication Process of CMOS Inverter (2/10)



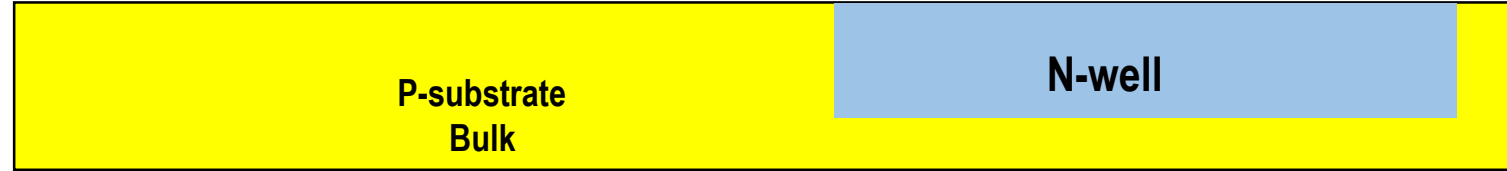
Step Description

Cross Section After applying the Steps



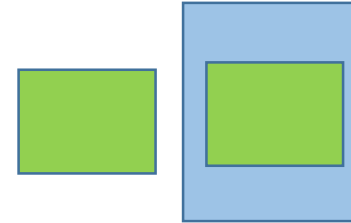
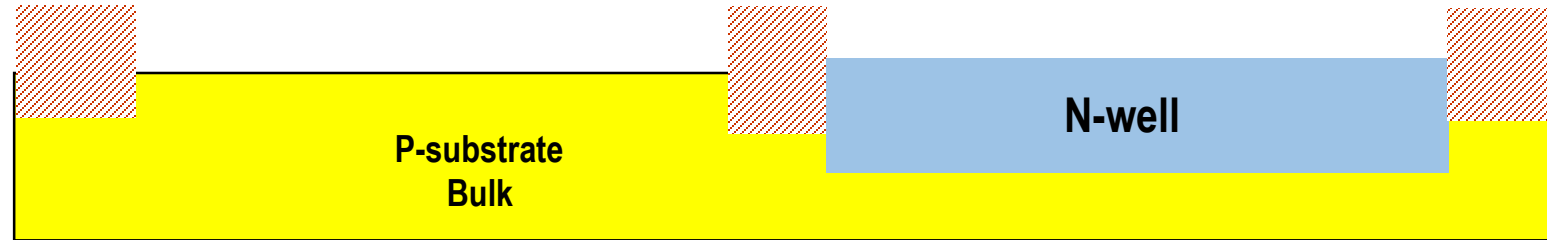
1- n-well creation

n-well mask, lithography,
N-well implant,
etching



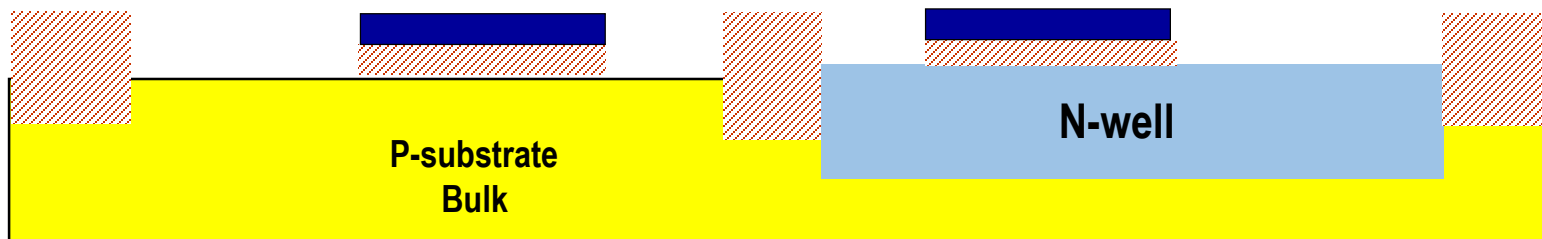
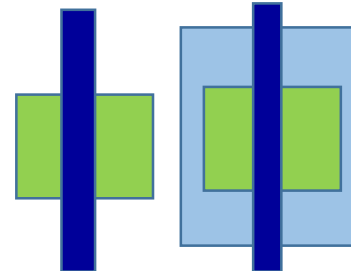
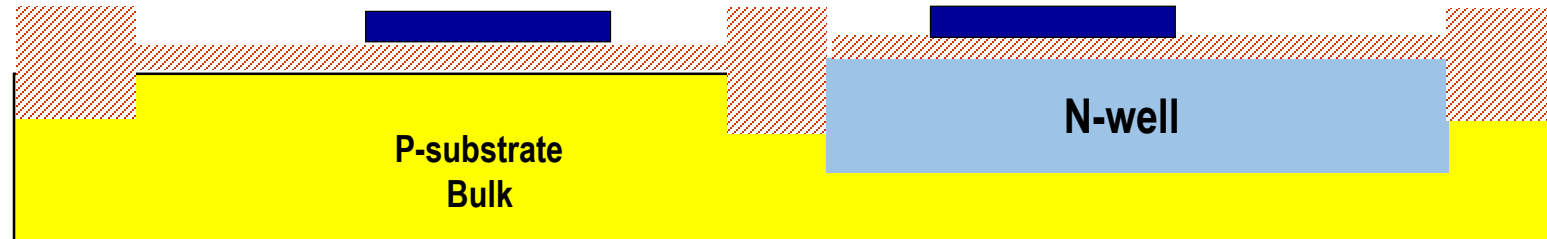
2- FOX creation

Active mask, lithography,
Field oxidation,
(FOX=Thick oxidation),
etching

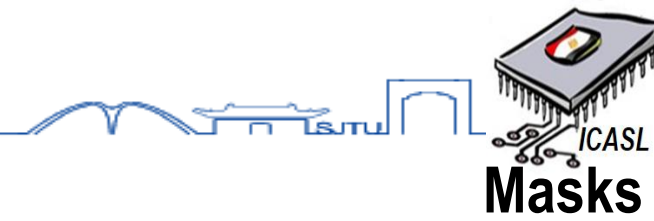


3- Gate creation

Thin oxidation, poly mask,
lithography,
poly deposition,
etching (thin oxidation)



Fabrication Process of CMOS Inverter (3/10)

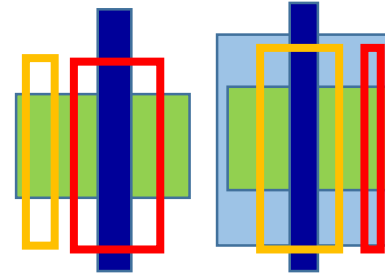
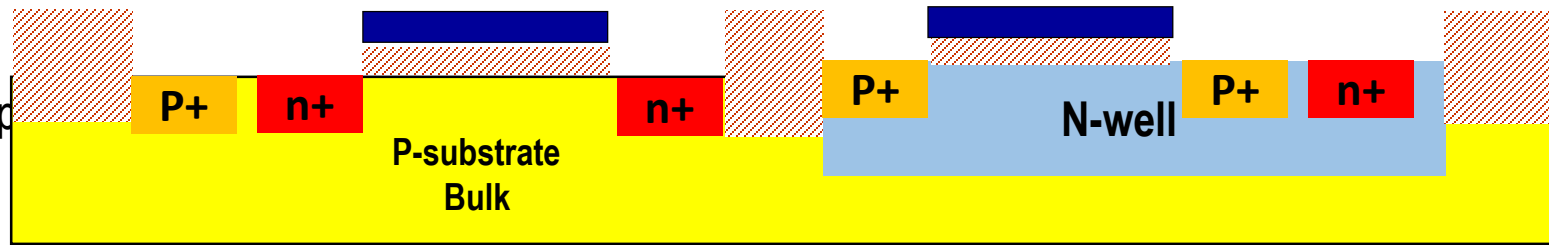


Step Description

Cross Section After applying the Steps

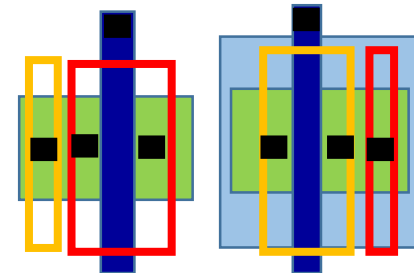
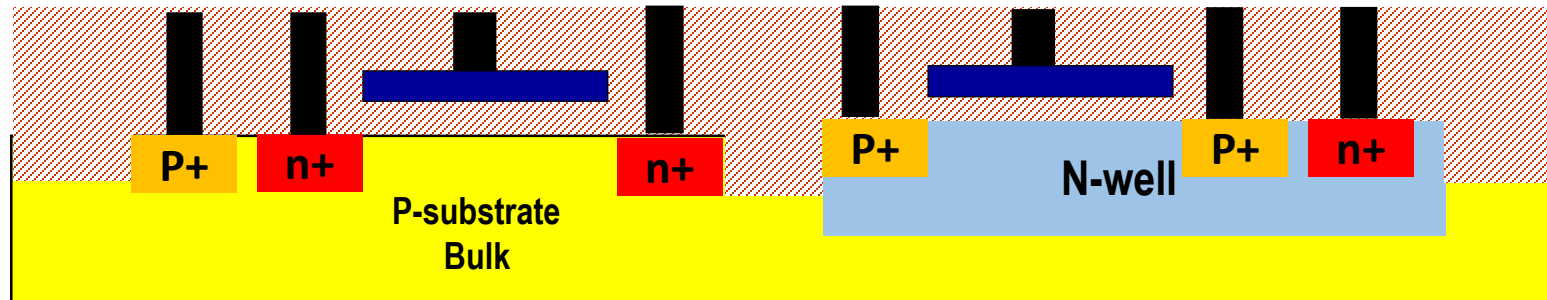
4- Drain/Source Creation

□ N-select Mask ,Lithography
Etching,
N+ drain/source diffusion
□ P-select Mask ,Lithography,
Etching,
P+ drain/source diffusion



5- Contact Creation

Fill oxide, polish
Active and poly
contact masks,
Lithography,
Contact etch, Metallization,
polish



Fabrication Process of CMOS Inverter (4/10)



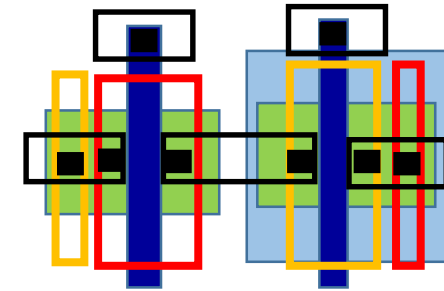
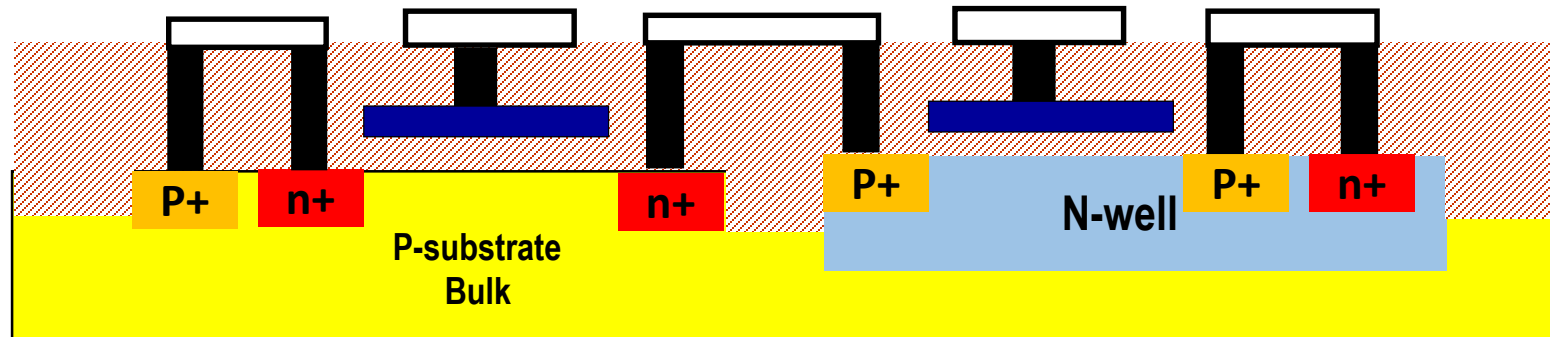
Step Description

Cross Section After applying the Steps

6- Metal 1 creation

M1 metallization,
M1 mask,
Lithography, Etching

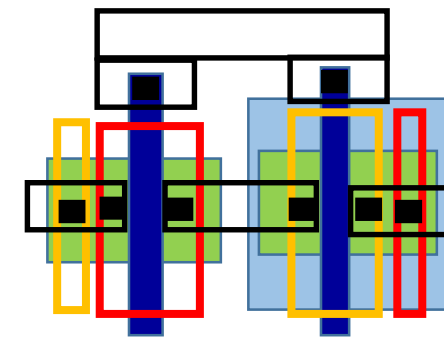
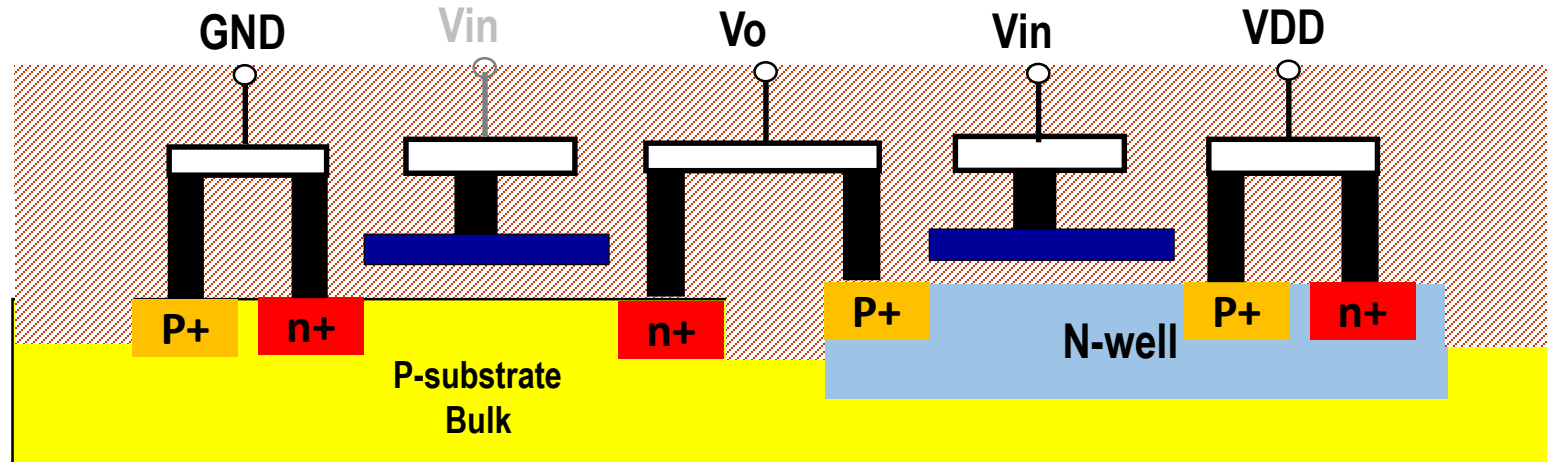
M1



Note:

If the provided technology has only 1X layer (Metal 1 layer)

M1



Fabrication Process of CMOS Inverter (5/10)



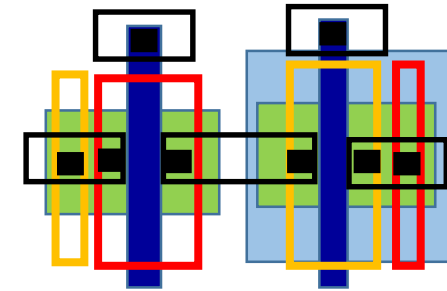
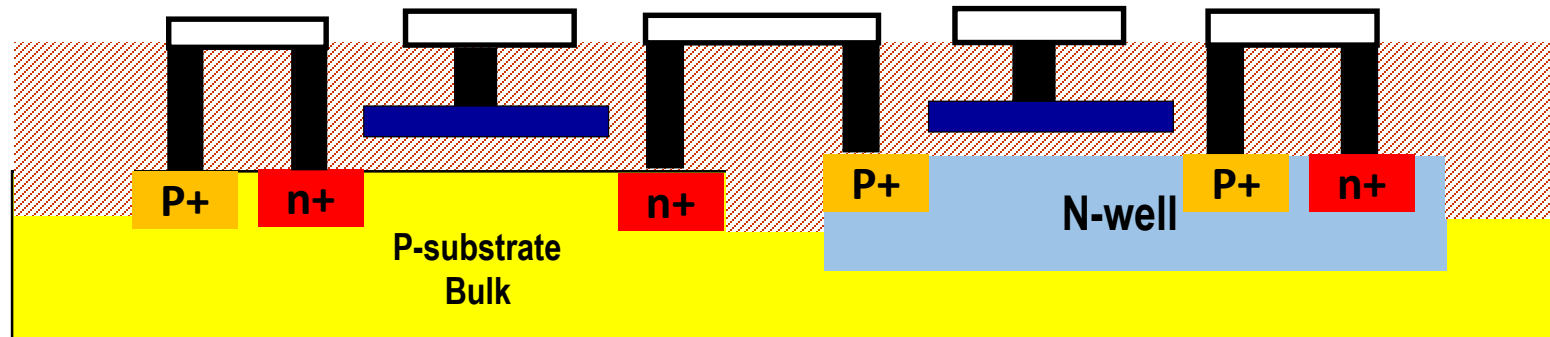
Step Description

Cross Section After applying the Steps

6- Metal 1 creation

M1 metallization,
M1 mask,
Lithography, Etching

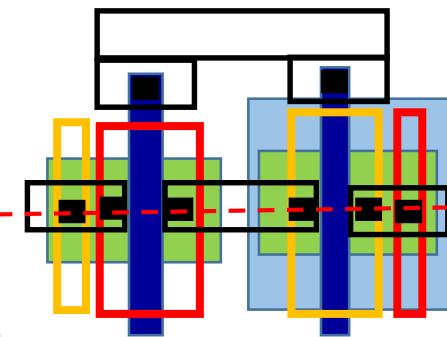
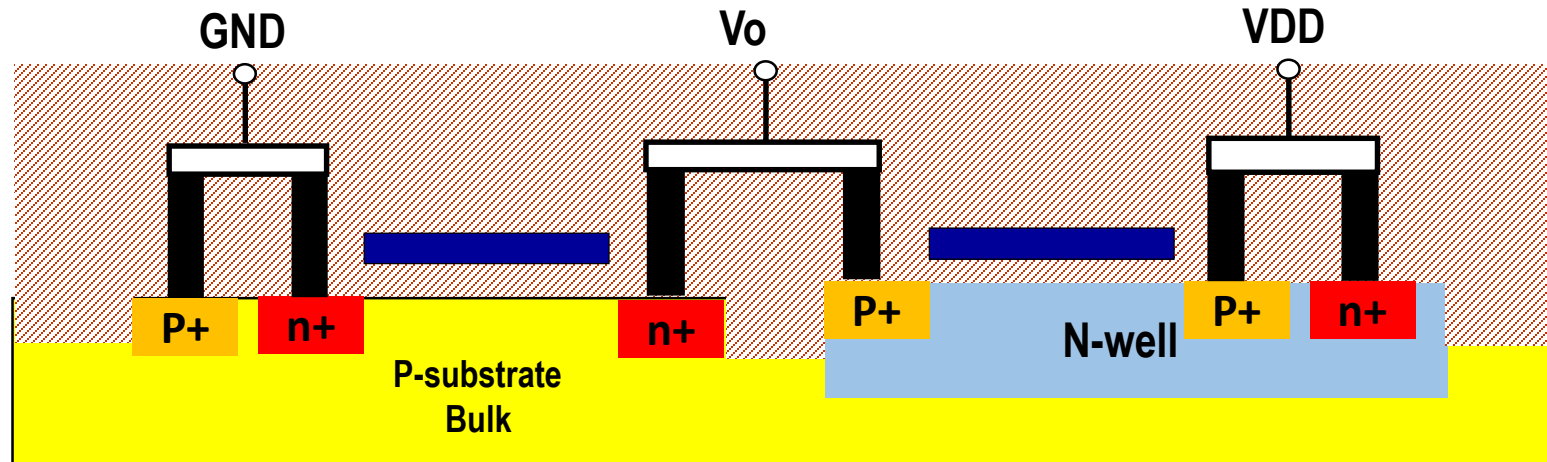
M1



Note:

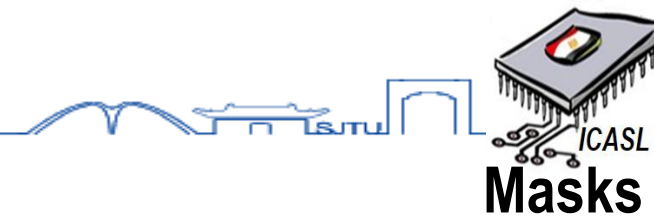
If the provided technology has only 1X layer (Metal 1 layer)

M1



Cross Section ←

Fabrication Process of CMOS Inverter (6/10)



Step Description

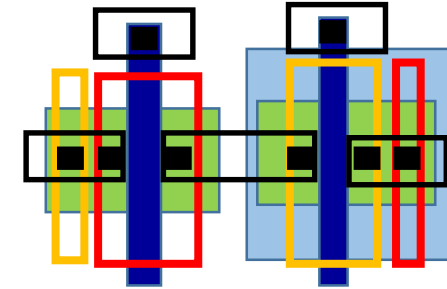
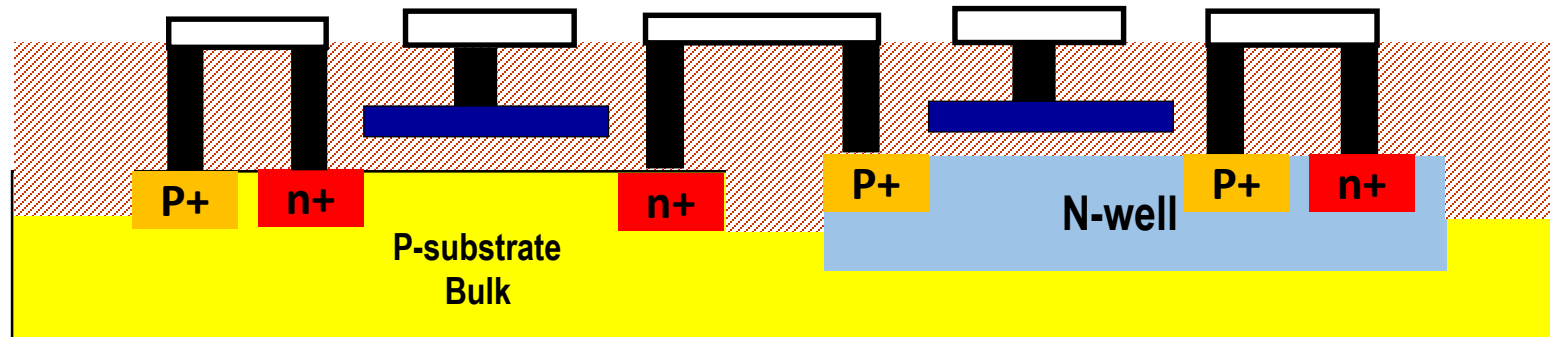
Cross Section After applying the Steps

Masks

6- Metal 1 creation

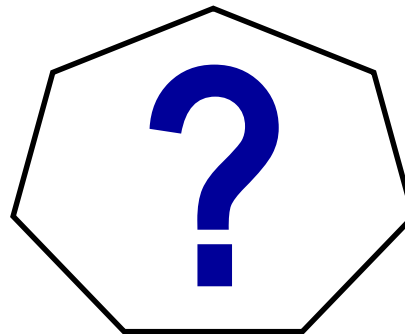
M1 metallization,
M1 mask,
Lithography, Etching

M1

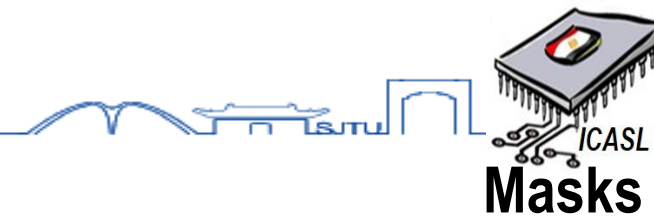


Note:

If the provided technology has only **2X** layers (Metal1 Layer+ Metal 2 Layer)



Fabrication Process of CMOS Inverter (7/10)



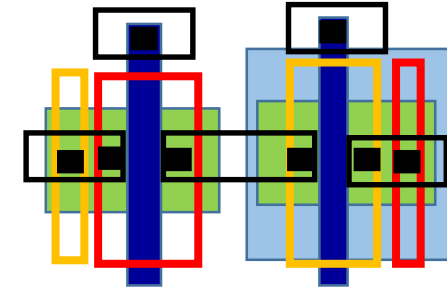
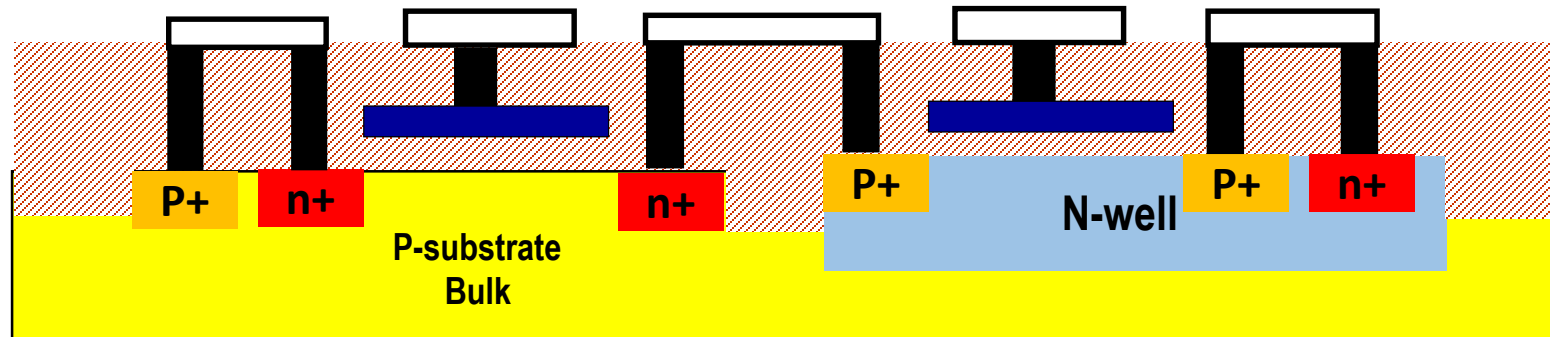
Step Description

Cross Section After applying the Steps

M1

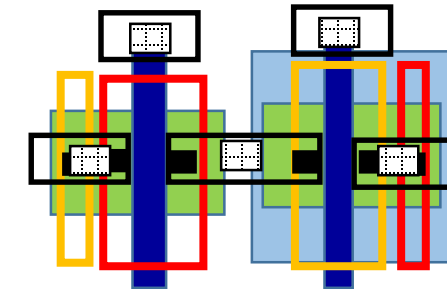
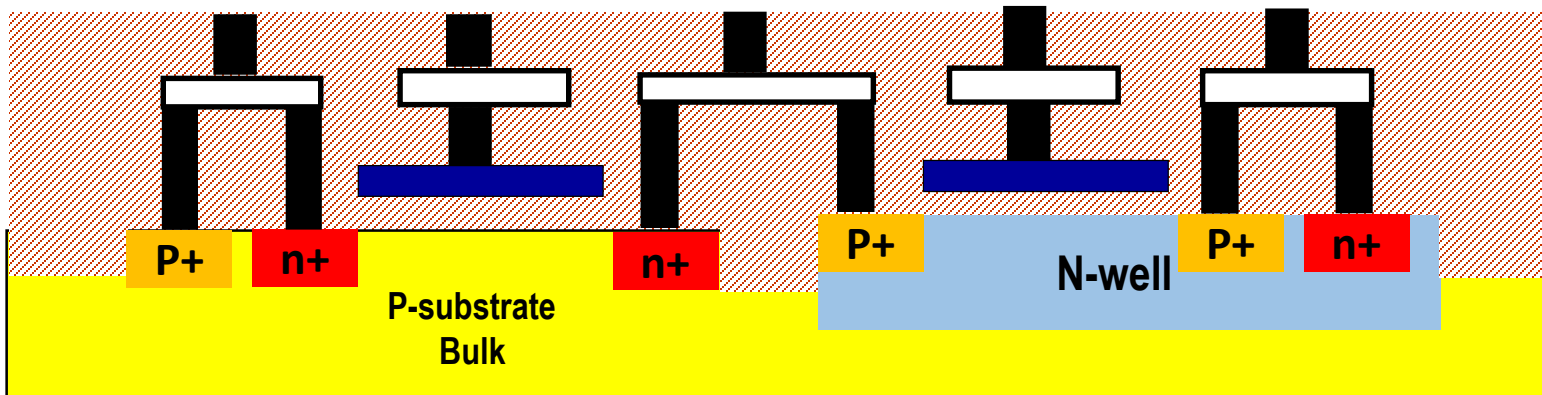
6- Metal 1 creation

M1 metallization,
M1 mask,
Lithography, Etching



7- Via 1 creation

Fill oxide,
Polish,
Via 1 mask,
Lithography,
Etching,



Fabrication Process of CMOS Inverter (8/10)



Step Description

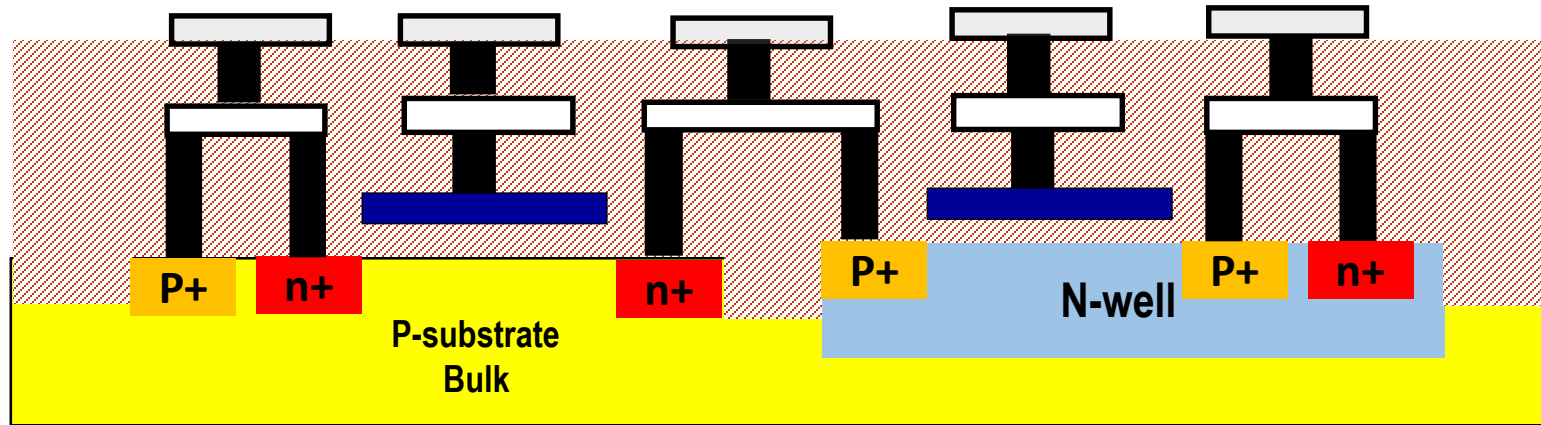
8- Metal 2 creation

M2 metallization,
M2 mask,
Lithography,
Etching

M2

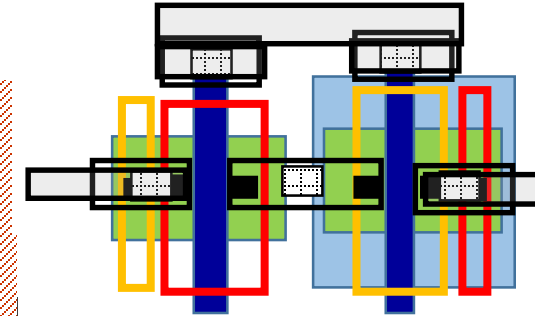
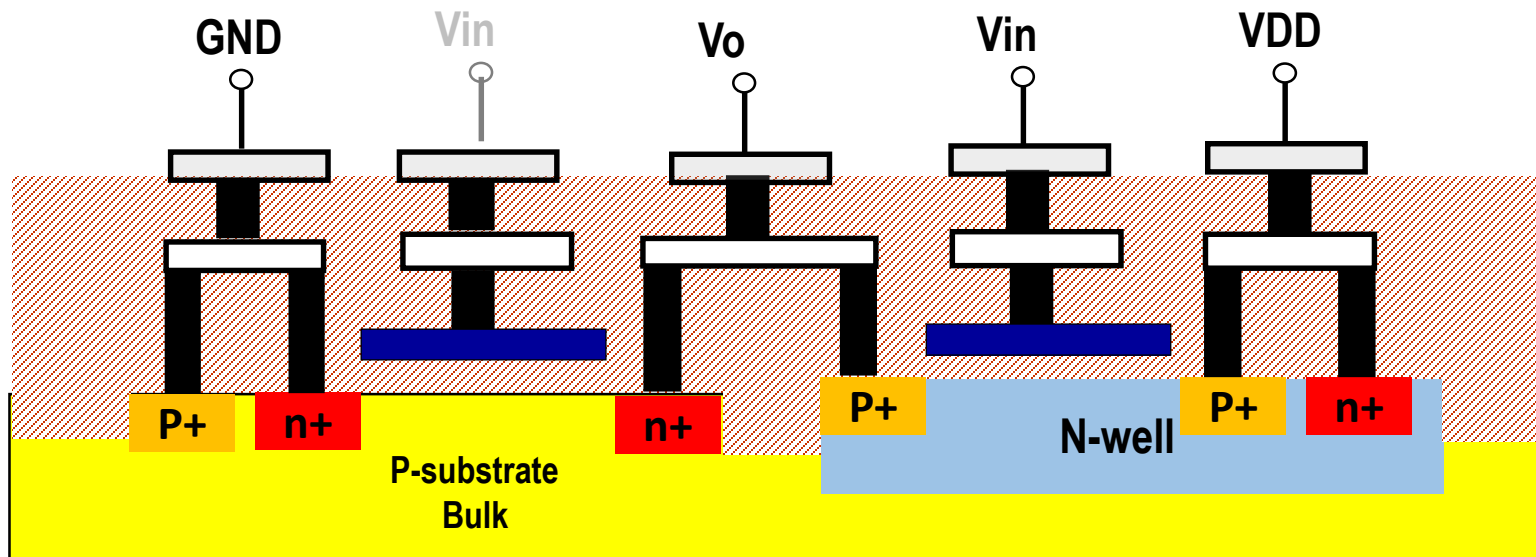
M1

Cross Section After applying the Steps



M2

M1



Masks

Fabrication Process of CMOS Inverter (9/10)



Step Description

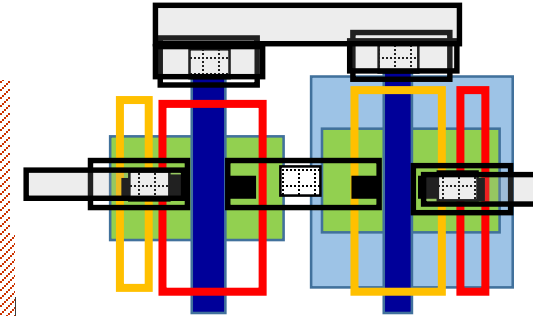
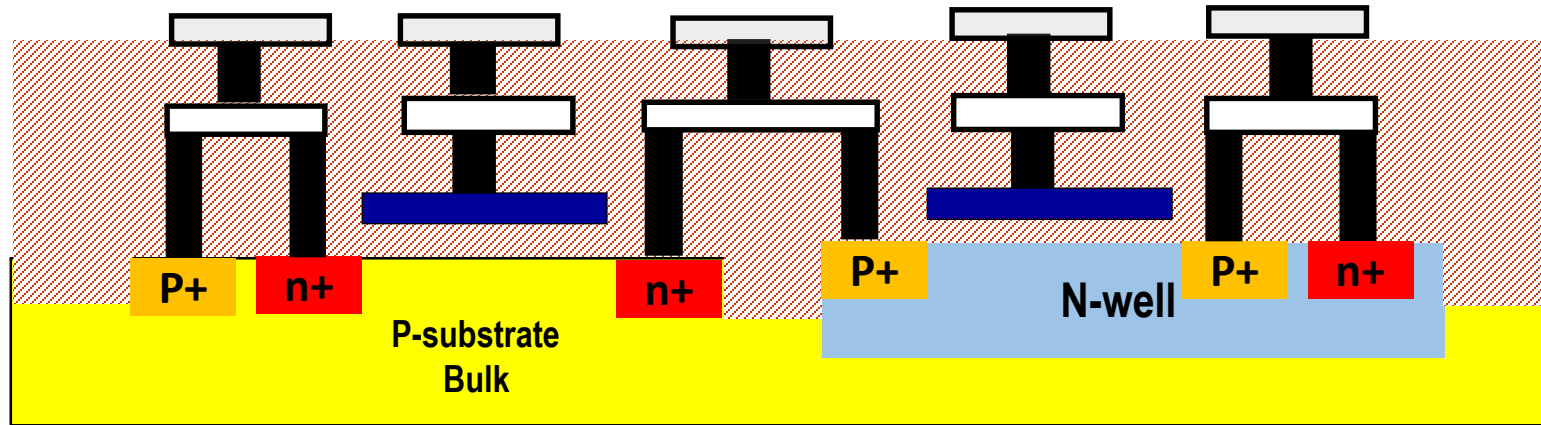
8- Metal 2 creation

M2 metallization,
M2 mask,
Lithography,
Etching

M2

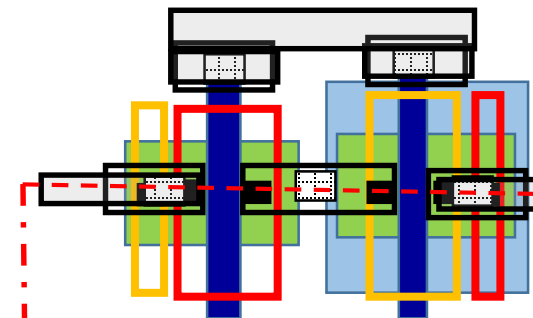
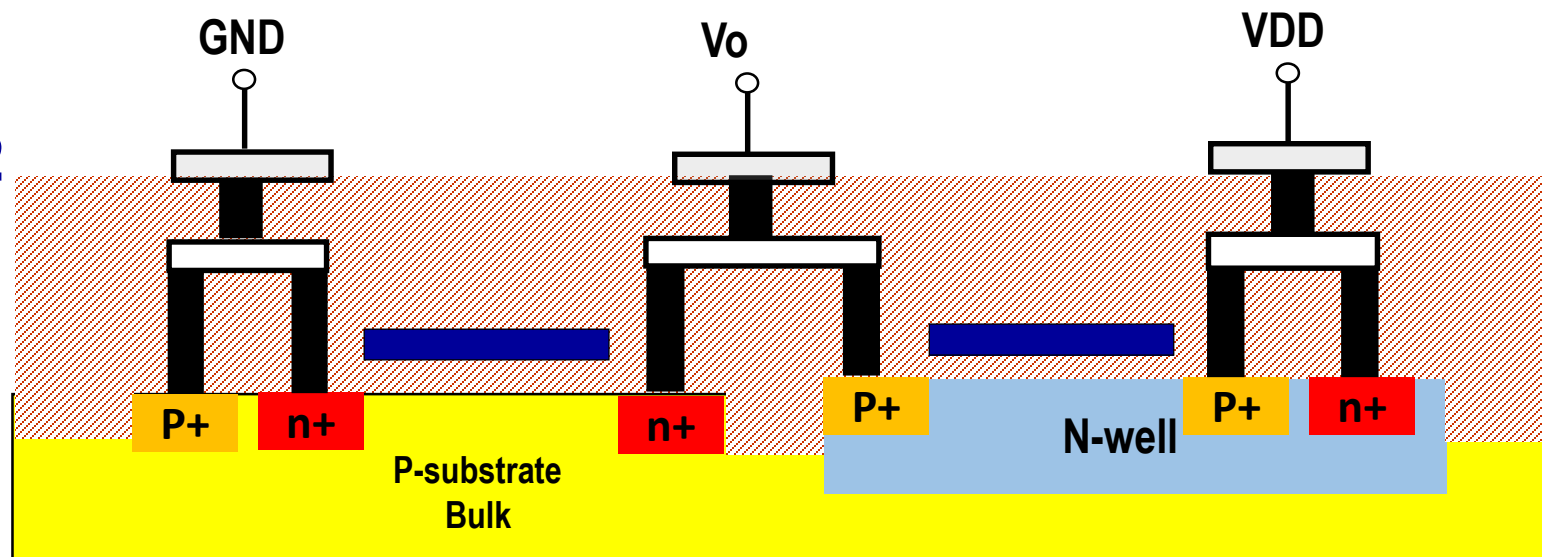
M1

Cross Section After applying the Steps



M2

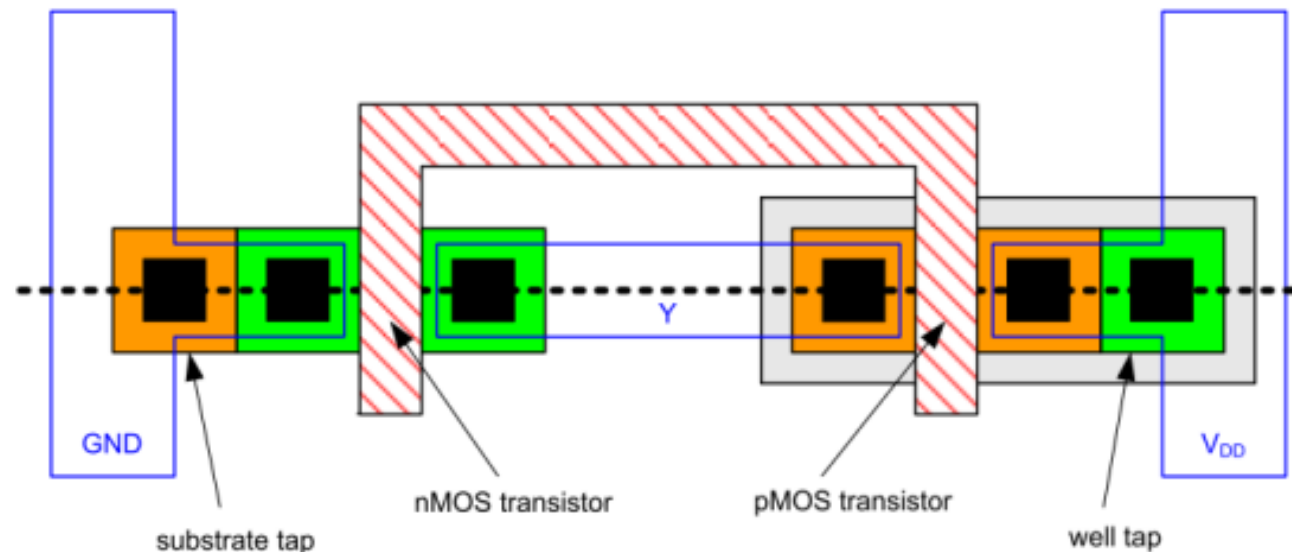
M1



Cross Section

Accumulated Masks are considered as **Layout** of the CMOS inverter

We will know how to draw the layout in details next lectures



Thanks!



- ☐ **Quiz/week (10 points) On Thursday at 10.00 Pm**
- ☐ **If you missed a Quiz, you would enhance it the next week**
- ☐ **Please don not miss writing Student name, Student ID, Student national ID.**
- ☐ **If you could not login the Quiz, please check it with the IT.**

Thanks!